Does an Accurate Prediction Come with a Reliable Explanation?

A Comparative Study of LIME and SHAP

Nguyễn Gia Hào 24125008
Phạm Hữu Nam 24125015
Nguyễn Hữu Phước 24125018

APCS – Faculty of Information Technology
VNUHCM - University of Science

CS420: Intro to AI – [S04] LIME & SHAP
Instructor: Dr. Nguyễn Ngọc Thảo

Explainable AI: key concepts

A model flags a tumour as *malignant*; the doctor asks *why*. An explanation method answers, but it is itself an algorithm with its own assumptions and settings.

Guiding question

Does a highly accurate prediction necessarily come with a reliable explanation?

Explainable AI: key concepts

A model flags a tumour as *malignant*; the doctor asks *why*. An explanation method answers, but it is itself an algorithm with its own assumptions and settings.

Guiding question

Does a highly accurate prediction necessarily come with a reliable explanation?

- **Interpretable vs explainable:** a model a human can follow directly (short tree, sparse linear model) vs explanations produced for a model that is not.

Explainable AI: key concepts

A model flags a tumour as *malignant*; the doctor asks *why*. An explanation method answers, but it is itself an algorithm with its own assumptions and settings.

Guiding question

Does a highly accurate prediction necessarily come with a reliable explanation?

- **Interpretable vs explainable:** a model a human can follow directly (short tree, sparse linear model) vs explanations produced for a model that is not.
- **Intrinsic vs post-hoc:** interpretable by construction vs a separate method explaining an already-trained model.

Explainable AI: key concepts

A model flags a tumour as *malignant*; the doctor asks *why*. An explanation method answers, but it is itself an algorithm with its own assumptions and settings.

Guiding question

Does a highly accurate prediction necessarily come with a reliable explanation?

- **Interpretable vs explainable:** a model a human can follow directly (short tree, sparse linear model) vs explanations produced for a model that is not.
- **Intrinsic vs post-hoc:** interpretable by construction vs a separate method explaining an already-trained model.
- **Local vs global:** explain one prediction $f(x)$ vs the model's overall behaviour.

Explainable AI: key concepts

A model flags a tumour as *malignant*; the doctor asks *why*. An explanation method answers, but it is itself an algorithm with its own assumptions and settings.

Guiding question

Does a highly accurate prediction necessarily come with a reliable explanation?

- **Interpretable vs explainable:** a model a human can follow directly (short tree, sparse linear model) vs explanations produced for a model that is not.
- **Intrinsic vs post-hoc:** interpretable by construction vs a separate method explaining an already-trained model.
- **Local vs global:** explain one prediction $f(x)$ vs the model's overall behaviour.
- **Model-specific vs model-agnostic:** use the model's internals vs only query f as a black box.

Explainable AI: key concepts

A model flags a tumour as *malignant*; the doctor asks *why*. An explanation method answers, but it is itself an algorithm with its own assumptions and settings.

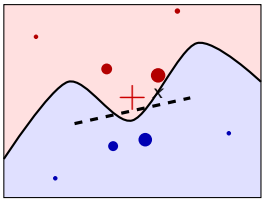
Guiding question

Does a highly accurate prediction necessarily come with a reliable explanation?

- **Interpretable vs explainable:** a model a human can follow directly (short tree, sparse linear model) vs explanations produced for a model that is not.
- **Intrinsic vs post-hoc:** interpretable by construction vs a separate method explaining an already-trained model.
- **Local vs global:** explain one prediction $f(x)$ vs the model's overall behaviour.
- **Model-specific vs model-agnostic:** use the model's internals vs only query f as a black box.

LIME (2016), SHAP (2017): post-hoc, local, model-agnostic. *Which features pushed this prediction up or down, and by how much?*

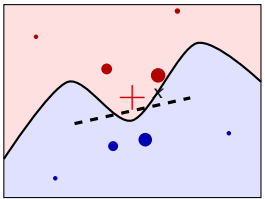
LIME: a simple model that is faithful *locally*



Shading: decision of f. Dot size: weight π_x. Dashed: local linear g.

Explain $f(x)$ by a simple g that only has to match f on *variations of* x (inputs similar to x).

LIME: a simple model that is faithful *locally*

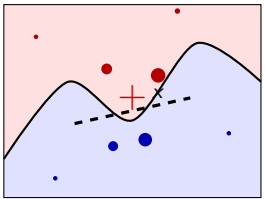


Shading: decision of f. Dot size: weight π_x. Dashed: local linear g.

Explain $f(x)$ by a simple g that only has to match f on *variations of* x (inputs similar to x).

Interpretable representation $x' = (1, \ldots, 1) \in \{0,1\}^{d'}$: bit j = "unit j present" (a word, a super-pixel, a value range). g sees x' and should give $g(x') \approx f(x)$.

LIME: a simple model that is faithful *locally*



Shading: decision of f. Dot size: weight π_x. Dashed: local linear g.

Explain $f(x)$ by a simple g that only has to match f on *variations of* x (inputs similar to x).

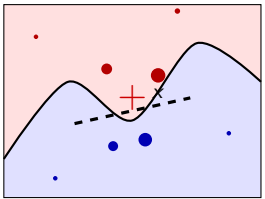
Interpretable representation $x' = (1, \ldots, 1) \in \{0,1\}^{d'}$: bit j = "unit j present" (a word, a super-pixel, a value range). g sees x' and should give $g(x') \approx f(x)$.

Variations: switch off random bits, map back with h_x; g sees z', f sees z:

$x' = (1,1,1,1) \xrightarrow{h_x}$ "the food was great"

$z' = (1,0,1,1) \xrightarrow{h_x}$ "the was great"

LIME: a simple model that is faithful *locally*



Shading: decision of f. Dot size: weight π_x. Dashed: local linear g.

Explain $f(x)$ by a simple g that only has to match f on *variations of* x (inputs similar to x).

Interpretable representation $x' = (1, \ldots, 1) \in \{0,1\}^{d'}$: bit j = "unit j present" (a word, a super-pixel, a value range). g sees x' and should give $g(x') \approx f(x)$.

Variations: switch off random bits, map back with h_x; g sees z', f sees z:

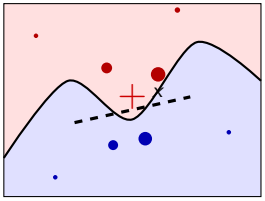
$x' = (1,1,1,1) \xrightarrow{h_x}$ "the food was great"

$z' = (1,0,1,1) \xrightarrow{h_x}$ "the was great"

Want $g(z') \approx f(z)$, weighted by proximity $\pi_x(z) = \exp(-D(x,z)^2/\sigma^2)$:

$$\xi(x) = \arg\min_{g \in G} \underbrace{\sum_{(z,z')} \pi_x(z)\left(f(z) - g(z')\right)^2}_{\mathcal{L}(f,g,\pi_x)\text{: faithful}} + \underbrace{\Omega(g)}_{\text{readable}} .$$

LIME: a simple model that is faithful *locally*



Shading: decision of f. Dot size: weight π_x. Dashed: local linear g.

Explain $f(x)$ by a simple g that only has to match f on *variations of* x (inputs similar to x).

Interpretable representation $x' = (1, \ldots, 1) \in \{0,1\}^{d'}$: bit j = "unit j present" (a word, a super-pixel, a value range). g sees x' and should give $g(x') \approx f(x)$.

Variations: switch off random bits, map back with h_x; g sees z', f sees z:

$x' = (1,1,1,1) \xrightarrow{h_x}$ "the food was great"

$z' = (1,0,1,1) \xrightarrow{h_x}$ "the was great"

Want $g(z') \approx f(z)$, weighted by proximity $\pi_x(z) = \exp(-D(x,z)^2/\sigma^2)$:

$$\xi(x) = \arg\min_{g \in G} \underbrace{\sum_{(z,z')} \pi_x(z)\big(f(z) - g(z')\big)^2}_{\mathcal{L}(f,g,\pi_x)\text{: faithful}} + \underbrace{\Omega(g)}_{\text{readable}} .$$

In LIME: sparse linear $g(z') = w_g \cdot z'$; $\Omega(g) = \infty$ if more than K weights are non-zero, else 0.

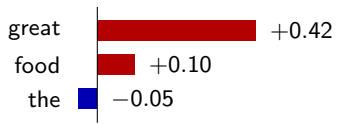
LIME: output and limitations

In practice: sample N variations (N calls to f), weight them by π_x, fit g with "K-LASSO" (Lasso picks K units, then weighted least squares on them).

LIME: output and limitations

In practice: sample N variations (N calls to f), weight them by π_x, fit g with "K-LASSO" (Lasso picks K units, then weighted least squares on them).

Output: K signed weights. $w_j > 0$ supports the class, $w_j < 0$ opposes it.

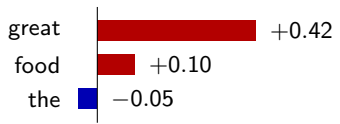


"the food was great", class *positive*, $K = 3$ (illustrative numbers).

LIME: output and limitations

In practice: sample N variations (N calls to f), weight them by π_x, fit g with "K-LASSO" (Lasso picks K units, then weighted least squares on them).

Output: K signed weights. $w_j > 0$ supports the class, $w_j < 0$ opposes it.



"the food was great", class *positive*, $K = 3$ (illustrative numbers).

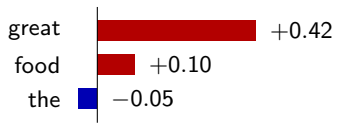
Limitations

- No local accuracy: $g(x')$ need not equal $f(x)$.

LIME: output and limitations

In practice: sample N variations (N calls to f), weight them by π_x, fit g with "K-LASSO" (Lasso picks K units, then weighted least squares on them).

Output: K signed weights. $w_j > 0$ supports the class, $w_j < 0$ opposes it.



"the food was great", class *positive*, $K = 3$ (illustrative numbers).

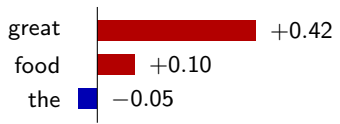
Limitations

- No local accuracy: $g(x')$ need not equal $f(x)$.
- Depends on the random variations, N, σ and the representation.

LIME: output and limitations

In practice: sample N variations (N calls to f), weight them by π_x, fit g with "K-LASSO" (Lasso picks K units, then weighted least squares on them).

Output: K signed weights. $w_j > 0$ supports the class, $w_j < 0$ opposes it.



"the food was great", class *positive*, $K = 3$ (illustrative numbers).

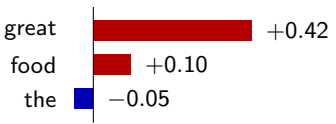
Limitations

- No local accuracy: $g(x')$ need not equal $f(x)$.
- Depends on the random variations, N, σ and the representation.
- Variations may be unrealistic; kernel and penalty are heuristic.

LIME: output and limitations

In practice: sample N variations (N calls to f), weight them by π_x, fit g with "K-LASSO" (Lasso picks K units, then weighted least squares on them).

Output: K signed weights. $w_j > 0$ supports the class, $w_j < 0$ opposes it.



"the food was great", class *positive*, $K = 3$ (illustrative numbers).

Limitations

- No local accuracy: $g(x')$ need not equal $f(x)$.
- Depends on the random variations, N, σ and the representation.
- Variations may be unrealistic; kernel and penalty are heuristic.

Next: SHAP

Which kernel and penalty make the explanation *unique*, with guaranteed properties?

SHAP: additive feature attribution

Same setting as LIME: a linear g on $z' \in \{0,1\}^M$,

$$g(z') = \phi_0 + \sum_{j=1}^{M} \phi_j z'_j,$$

$z'_j = 1$: feature j *known* (keeps x_j), $z'_j = 0$: *missing*.

SHAP: additive feature attribution

Same setting as LIME: a linear g on $z' \in \{0,1\}^M$,

$$g(z') = \phi_0 + \sum_{j=1}^{M} \phi_j z'_j,$$

$z'_j = 1$: feature j *known* (keeps x_j), $z'_j = 0$: *missing*.

Missing = averaged out over a background (reference rows b):

$$f_x(S) = \mathbb{E}[f(z) \mid z_S = x_S] \approx \mathbb{E}_b\big[f(x_S, b_{\bar{S}})\big].$$

Base value $\phi_0 = f_x(\varnothing)$: the average prediction.

SHAP: additive feature attribution

Same setting as LIME: a linear g on $z' \in \{0,1\}^M$,

$$g(z') = \phi_0 + \sum_{j=1}^{M} \phi_j z'_j,$$

$z'_j = 1$: feature j *known* (keeps x_j), $z'_j = 0$: *missing*.

Missing = averaged out over a background (reference rows b):

$$f_x(S) = \mathbb{E}[f(z) \mid z_S = x_S] \approx \mathbb{E}_b\big[f(x_S, b_{\bar{S}})\big].$$

Base value $\phi_0 = f_x(\varnothing)$: the average prediction.

Sign: $\phi_j > 0$ pushes the output above ϕ_0, $\phi_j < 0$ below.

SHAP: additive feature attribution

Same setting as LIME: a linear g on $z' \in \{0,1\}^M$,

$$g(z') = \phi_0 + \sum_{j=1}^{M} \phi_j z'_j,$$

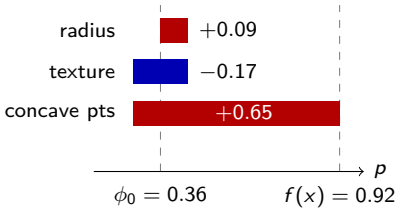
$z'_j = 1$: feature j *known* (keeps x_j), $z'_j = 0$: *missing*.

Missing = averaged out over a background (reference rows b):

$$f_x(S) = \mathbb{E}[f(z) \mid z_S = x_S] \approx \mathbb{E}_b\big[f(x_S, b_{\bar{S}})\big].$$

Base value $\phi_0 = f_x(\varnothing)$: the average prediction.

Sign: $\phi_j > 0$ pushes the output above ϕ_0, $\phi_j < 0$ below.



Ours: 3-feature logistic regression on Breast Cancer; output = probability of malignant.

$0.36 + 0.09 - 0.17 + 0.65 = 0.92$.

Shapley values: the unique split with three properties

- **Local accuracy:** $f(x) = \phi_0 + \sum_j \phi_j$ (LIME does not guarantee this).

Shapley values: the unique split with three properties

- **Local accuracy:** $f(x) = \phi_0 + \sum_j \phi_j$ (LIME does not guarantee this).
- **Missingness:** a unit absent from x ($x'_j = 0$) gets $\phi_j = 0$.

Shapley values: the unique split with three properties

- **Local accuracy:** $f(x) = \phi_0 + \sum_j \phi_j$ (LIME does not guarantee this).
- **Missingness:** a unit absent from x ($x'_j = 0$) gets $\phi_j = 0$.
- **Consistency:** if j's marginal contributions never decrease, ϕ_j does not decrease.

Shapley values: the unique split with three properties

- **Local accuracy:** $f(x) = \phi_0 + \sum_j \phi_j$ (LIME does not guarantee this).
- **Missingness:** a unit absent from x ($x'_j = 0$) gets $\phi_j = 0$.
- **Consistency:** if j's marginal contributions never decrease, ϕ_j does not decrease.

Theorem 1: only the Shapley values satisfy all three

$$\phi_j = \sum_{S \subseteq \{1,\dots,M\} \setminus \{j\}} \frac{|S|!\,(M - |S| - 1)!}{M!} \left[f_x(S \cup \{j\}) - f_x(S) \right]$$

Shapley values: the unique split with three properties

- **Local accuracy:** $f(x) = \phi_0 + \sum_j \phi_j$ (LIME does not guarantee this).
- **Missingness:** a unit absent from x ($x'_j = 0$) gets $\phi_j = 0$.
- **Consistency:** if j's marginal contributions never decrease, ϕ_j does not decrease.

Theorem 1: only the Shapley values satisfy all three

$$\phi_j = \sum_{S \subseteq \{1,\dots,M\} \setminus \{j\}} \frac{|S|!\,(M - |S| - 1)!}{M!} \left[f_x(S \cup \{j\}) - f_x(S) \right]$$

Reveal features in a random order: ϕ_j = average jump of the output when j is revealed.

Example: score 0 (no symptom), 5 (fever or cough alone), 2 (both):
$\phi_{\text{fever}} = \frac{1}{2}[(5-0) + (2-5)] = 1 = \phi_{\text{cough}}, \;\; 0 + 1 + 1 = 2 = f(x).$

Computing SHAP values

Kernel SHAP (Theorem 2): LIME's objective gives exactly the Shapley values with

$$\Omega(g) = 0, \qquad \pi_x(z') = \frac{M-1}{\binom{M}{|z'|}\,|z'|\,(M-|z'|)}, \qquad \text{constraints } \phi_0 = f_x(\varnothing),\ \sum_j \phi_j = f(x) - \phi_0.$$

Computing SHAP values

Kernel SHAP (Theorem 2): LIME's objective gives exactly the Shapley values with

$$\Omega(g) = 0, \qquad \pi_x(z') = \frac{M-1}{\binom{M}{|z'|}\,|z'|\,(M-|z'|)}, \qquad \text{constraints } \phi_0 = f_x(\varnothing),\ \sum_j \phi_j = f(x) - \phi_0.$$

	LIME	Kernel SHAP
Weight of a variation	proximity, width σ by hand	coalition size only, no width
Complexity Ω	at most K weights	none
Constraints	none	local accuracy
Result	heuristic weights	Shapley values (enough samples)

Computing SHAP values

Kernel SHAP (Theorem 2): LIME's objective gives exactly the Shapley values with

$$\Omega(g) = 0, \qquad \pi_x(z') = \frac{M-1}{\binom{M}{|z'|}\,|z'|\,(M-|z'|)}, \qquad \text{constraints } \phi_0 = f_x(\varnothing),\ \sum_j \phi_j = f(x) - \phi_0.$$

	LIME	Kernel SHAP
Weight of a variation	proximity, width σ by hand	coalition size only, no width
Complexity Ω	at most K weights	none
Constraints	none	local accuracy
Result	heuristic weights	Shapley values (enough samples)

Model-specific: Linear SHAP $\phi_j = w_j\,(x_j - \mathbb{E}[x_j])$ in closed form; Tree SHAP exact for tree ensembles.

Computing SHAP values

Kernel SHAP (Theorem 2): LIME's objective gives exactly the Shapley values with

$$\Omega(g) = 0, \qquad \pi_x(z') = \frac{M-1}{\binom{M}{|z'|}\,|z'|\,(M-|z'|)}, \qquad \text{constraints } \phi_0 = f_x(\varnothing),\ \textstyle\sum_j \phi_j = f(x) - \phi_0.$$

	LIME	Kernel SHAP
Weight of a variation	proximity, width σ by hand	coalition size only, no width
Complexity Ω	at most K weights	none
Constraints	none	local accuracy
Result	heuristic weights	Shapley values (enough samples)

Model-specific: Linear SHAP $\phi_j = w_j\,(x_j - \mathbb{E}[x_j])$ in closed form; Tree SHAP exact for tree ensembles.

Caveats: sampling noise for non-linear f; ϕ_0 and all ϕ_j depend on the background; independence creates unrealistic inputs for correlated features.

References

1. M. T. Ribeiro, S. Singh, C. Guestrin. "Why Should I Trust You?": Explaining the Predictions of Any Classifier. *KDD*, 2016.
2. S. M. Lundberg, S.-I. Lee. A Unified Approach to Interpreting Model Predictions. *NeurIPS*, 2017.
3. S. M. Lundberg et al. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2:56–67, 2020.
4. Explainable artificial intelligence: A survey of needs, techniques, applications, and future direction. *Neurocomputing*, 599:128111, 2024.